KYAMBOGO UNIVERSITY

DEPARTMENT OF LIBRARY AND INFORMATION SCIENCE

SCHOOL OF COMPUTING AND INFORMATION SCIENCE

YEAR: TWO

SEMESTER: II

COUSRE UNIT: INFORMATION RETRIEVAL

COUSRE CODE: ALIS 2205

LECTURERS NAME: DR. NAMBOBI OLIVIA

GROUP 4

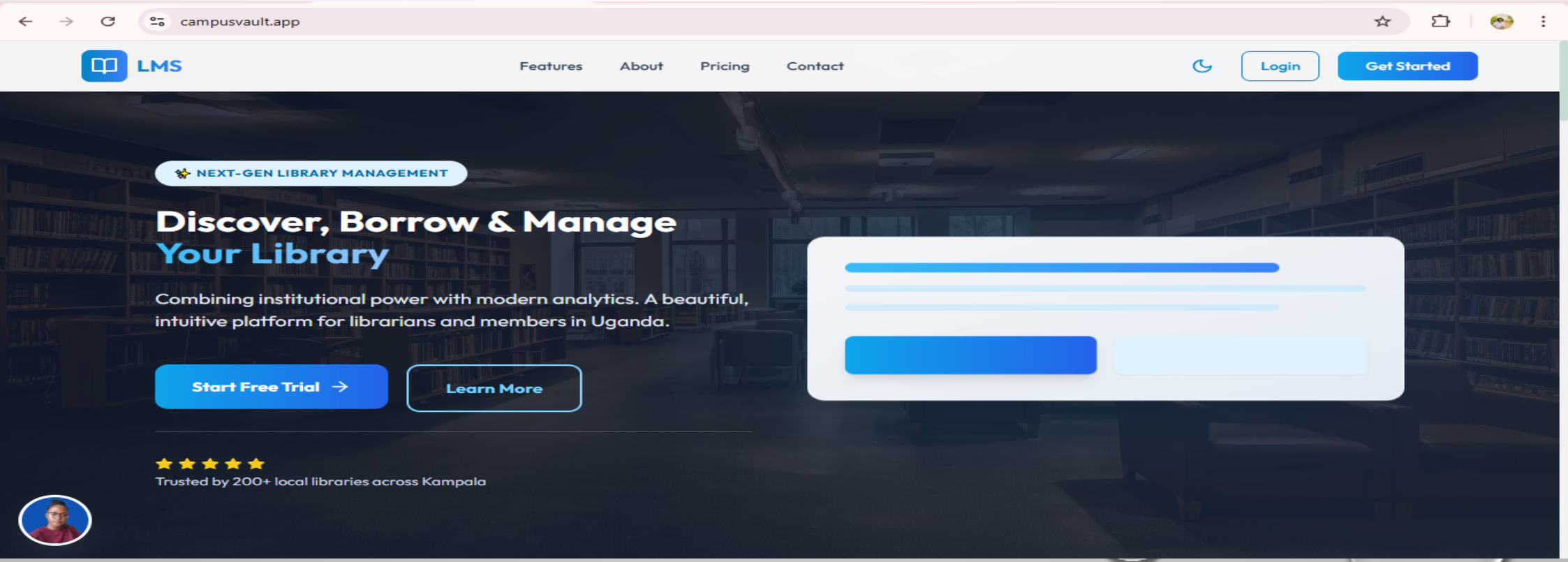
1. OWINO NORBERT DEOGRATIAS – 24/U/LID/18602/PD
2. KIBUUKA MATHIAS – 24/U/LID/06033/PD
3. SSIMBWA ALVIN – 24/U/LID/11890/PD
4. LULE STEVEN – 24/U/LID/06815/PD
5. NAKANWAGI DEBORAH – 24/U/LID/08699/PD
6. MBAZIIRA SHAFIQ – 24/U/LID/07143/PD
7. ACHOM HANIFAH OMODING – 24/U/LIE/02077/PE
8. MBAWADDE DOREEN – 24/U/LID/07136/PD
9. KYAZZE VICTORIA – 24/U/LID/06593/PD
10. KATAANO VIVIAN LYNNAH – 24/U/LID/05699/PD
11. SSENYONGA LUCY – 24/U/LID/11850/PD
12. AUMA IMMUCULATE PEACE – 24/U/LID/03961/PD

QUESTIONS

a) Design a RETRIEVAL SYSTEM ie. databases, integrated library systems in any software of your choice your can use trial versions on the network

b) Design a retrieval system approaches either technology centred and user centred approaches

DESIGNING A RETRIEVAL SYSTEM USING: Library Management System Database.



INTRODUCTION

- A retrieval system is a system designed to collect, store, organize and retrieve information when users search for it. They are usually integrated library systems. . It allows users to find relevant information from large collections of data, such as text documents, databases, or digital libraries.
- In the long run, , libraries require efficient database systems and integrated library systems to organize, store, retrieve, and disseminate information effectively.
- The main purpose is to match the user's information need with the stored information efficiently and accurately.
- **Example;**

➤ Library catalogue eg. (OPAC)

➤ Digital repository

➤ Online databases

Importance of a retrieval system

- **Fast access to information.** Allows users to search and obtain information quickly using keywords, titles, authors, or subjects.
- **Efficient organization of information**. Organizes information systematically using metadata, classification systems, indexing hence making information easy to locate and manage.
- **Improved accuracy.** Reduce human errors in searching and record keeping.
- **Remote access.** Allow users to search online through OPAC or digital platforms.

- **Time-Saving:** Instead of manually searching through shelves or catalogs, users can retrieve information within seconds using search queries.
- **Improved Resource Management:** Libraries can track usage patterns, understand popular topics, and manage collections efficiently

The link that takes you to our retrieval system

https://campusvault.app/

QUESTION 4B

Design Retrieval system approaches either technology- centered OR user- centered approach

USER CENTRED APPROACH

- A user-centred approach focuses primarily on the needs, behaviours, and experiences of users when interacting with an information retrieval system. The system is designed to make information searching simple and accessible.

There are various user centered databases that include the following

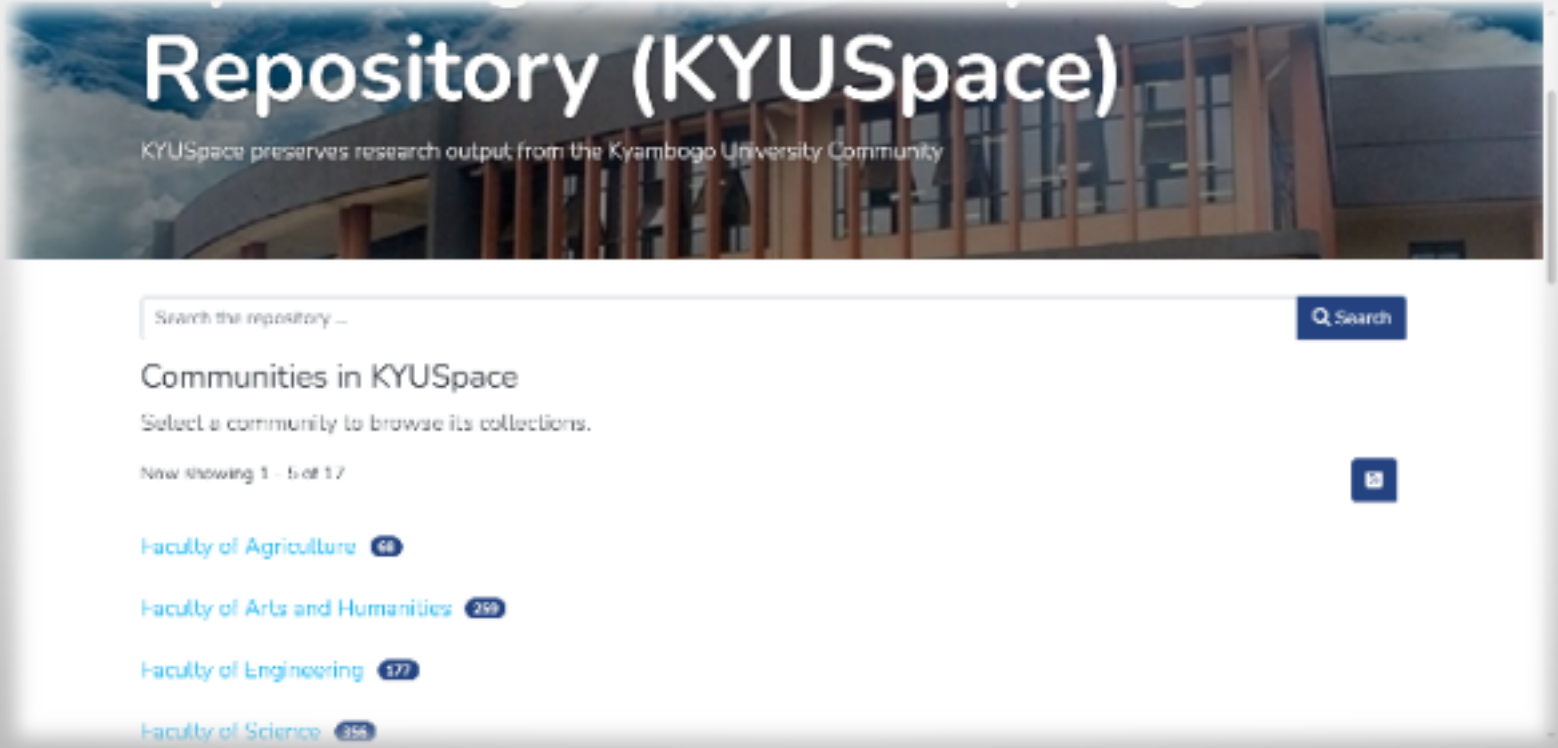
Kyuspace, Proquest, EBSCOhost, Google scholar and so many others.

However, this study will focus on kyuspace, the institutional repository of Kyambogo university because of its user-centered database for accessing academic research materials such as theses, dissertations, and projects.

FEATURES OF USER CENTRED

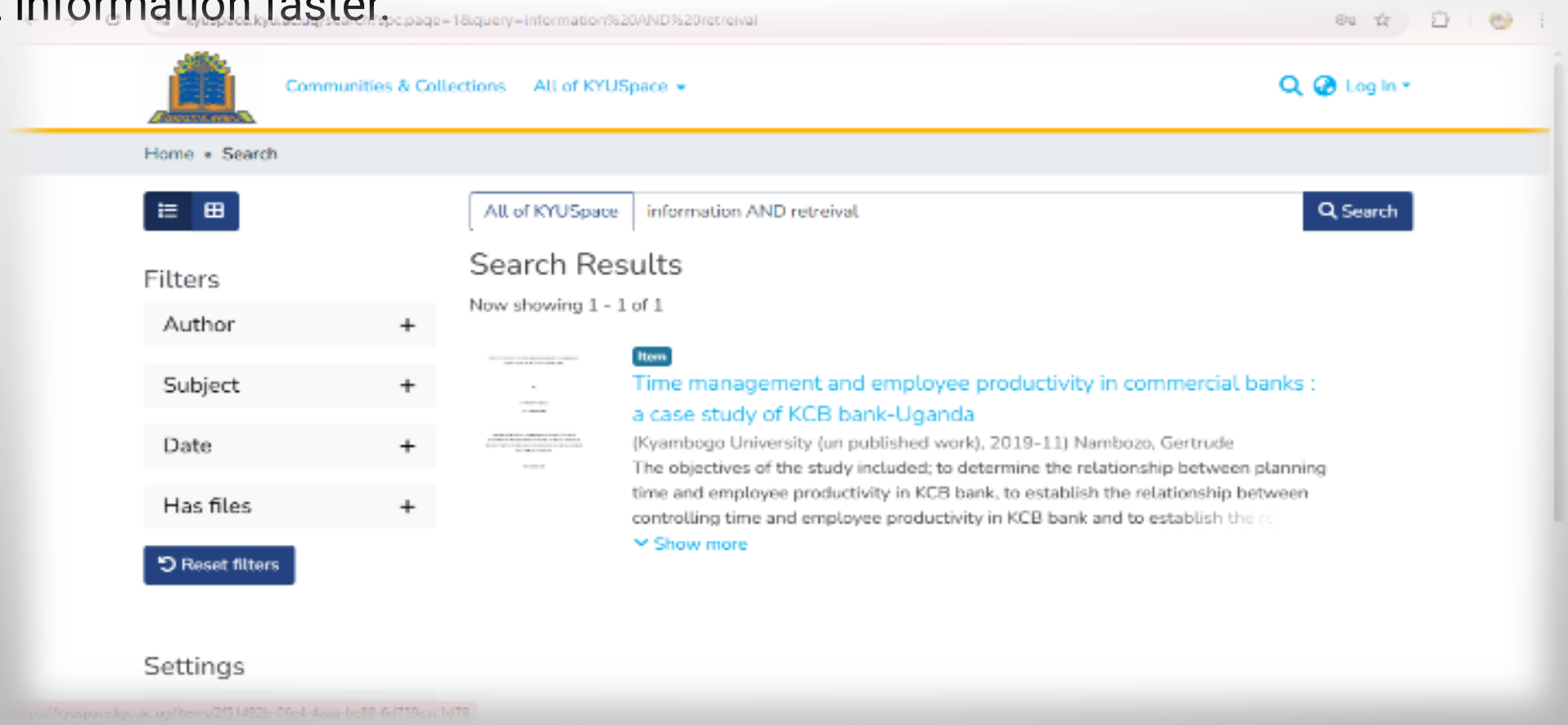
- **Searchability**

is the ability to search for information using keywords, titles, authors, or subjects. In KYU space, users can type keywords in the search bar to find specific research materials stored in the repository.



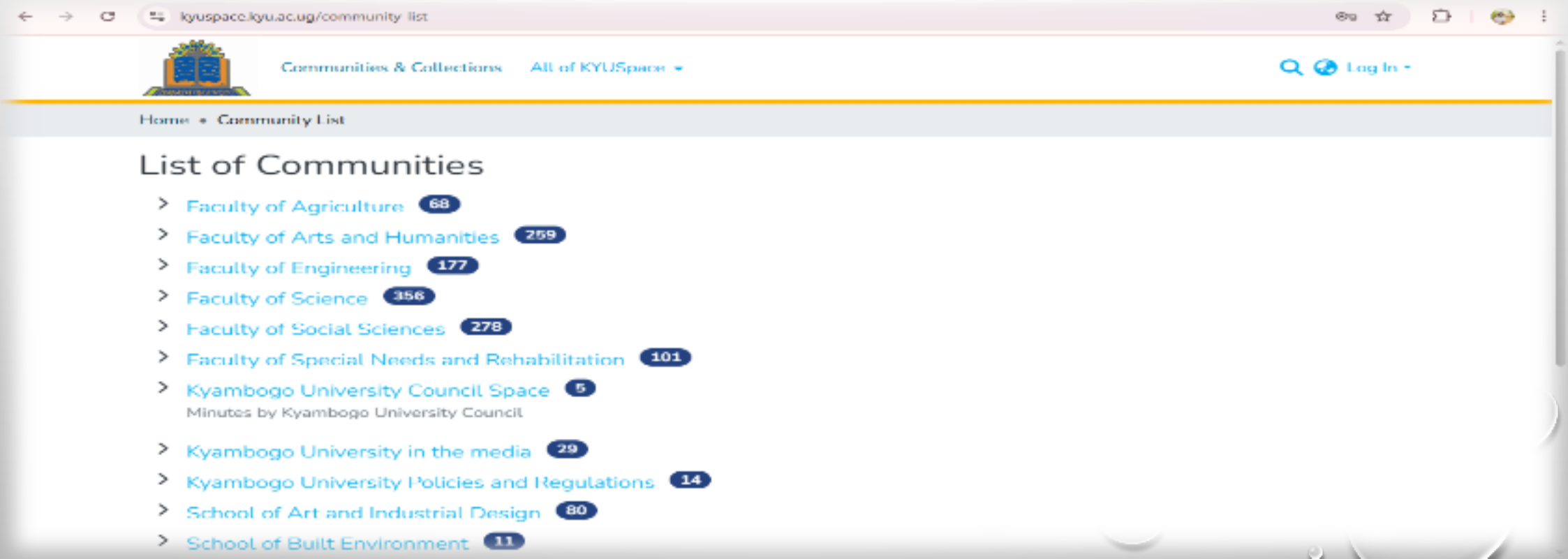
Filtering / advanced search filtering

Allows users to narrow down search results. In KYU space, users can filter results by year, author, subject, department, or type of document, which helps them find relevant information faster.



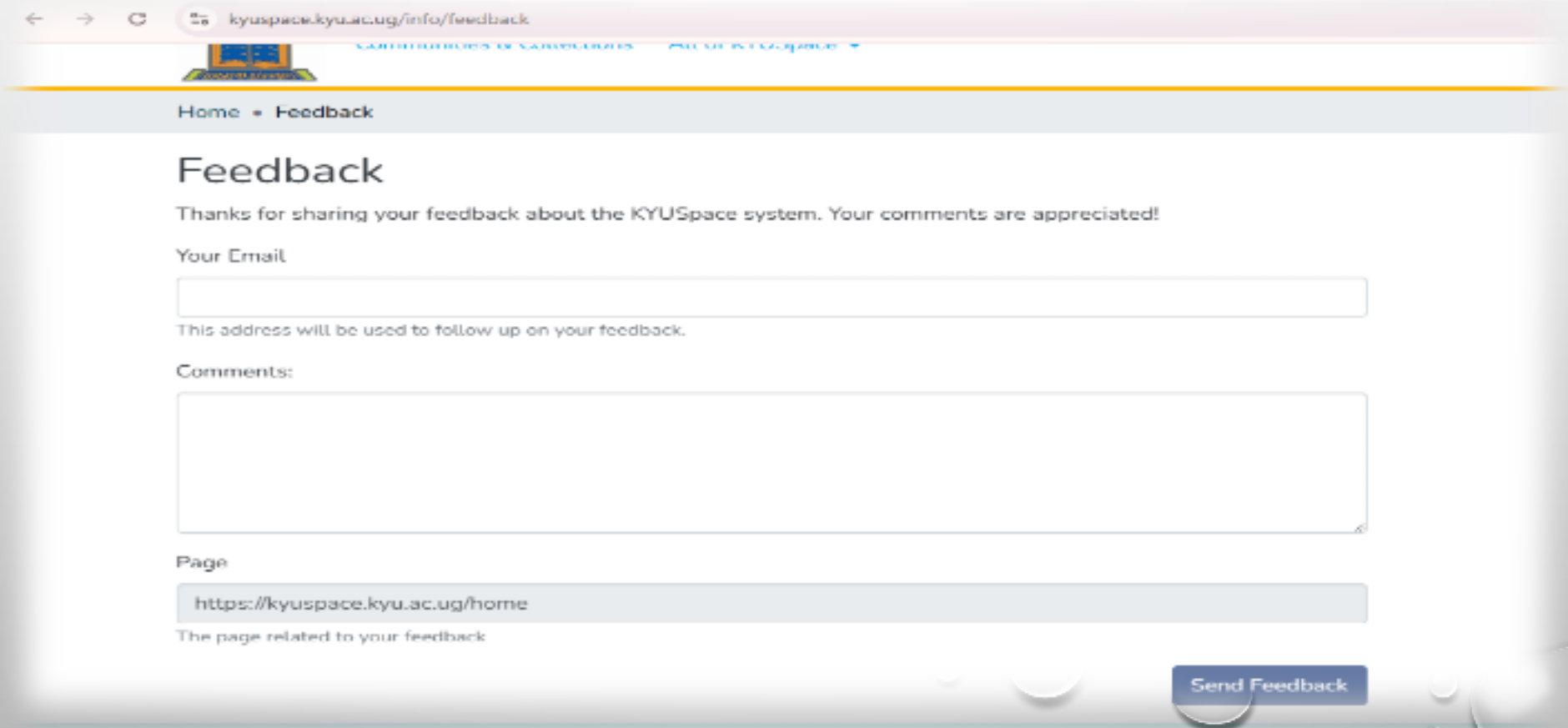
Interactive features

Help users interact with the system to improve their search experience. In Kyuspace, users can browse collections, view abstracts, and navigate through different departments and research categories.



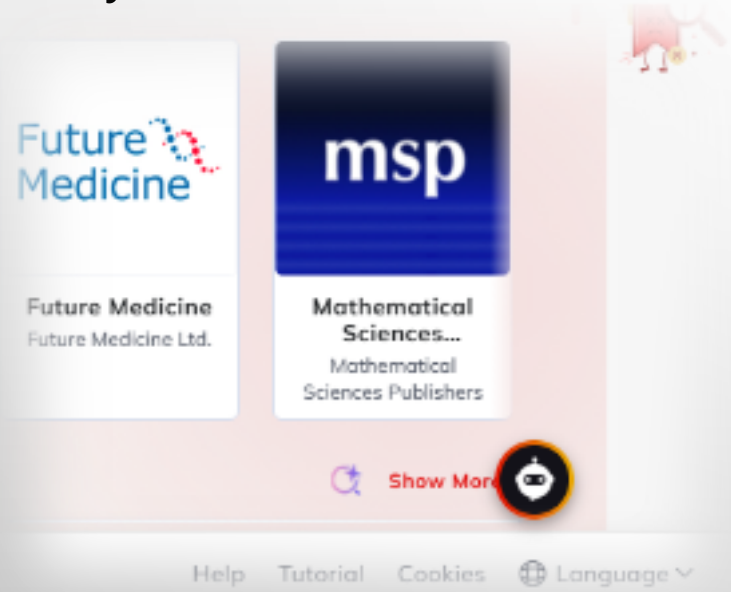
Feedback

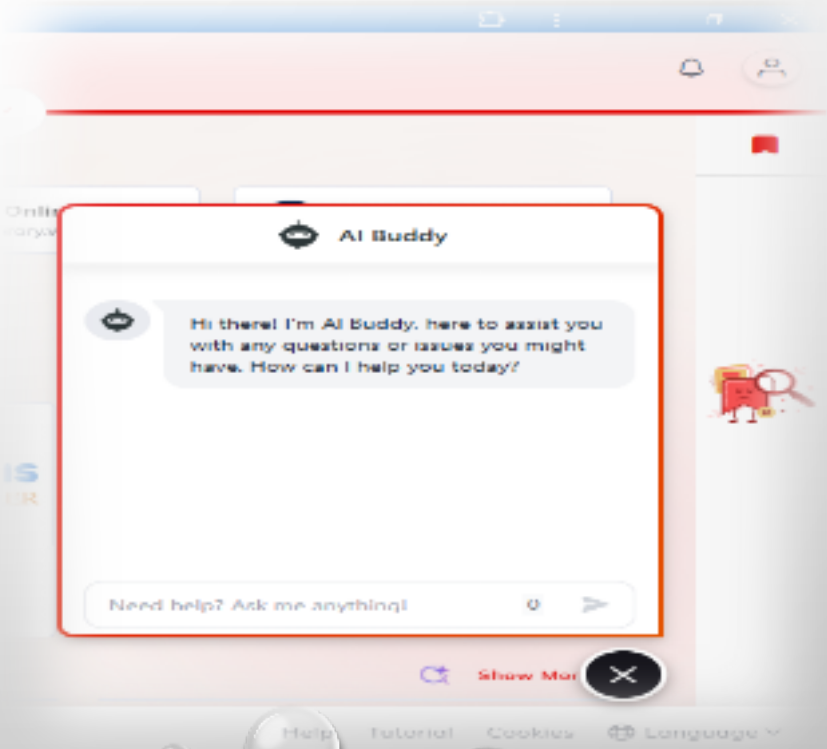
This allows users to communicate with the system administrators or librarians. For example, users can report problems, request assistance, or suggest improvements related to accessing research materials in Kyuspace.



Chatbox

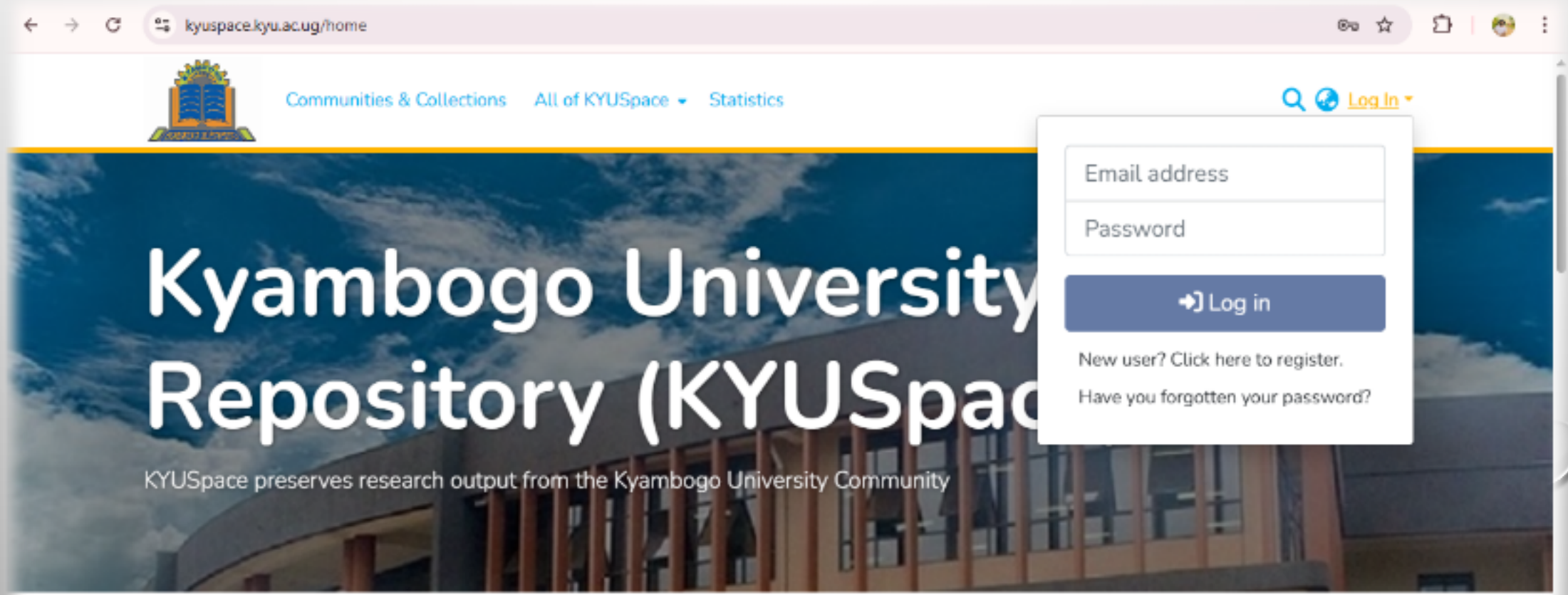
Helps users get instant assistance from librarians or support staff. If available, it allows students to ask questions about finding research materials or using the repository.





Personalization

This allows the users to create accounts and manage their own information activities. In Kyu space, researchers or students can log in, submit their research work, and manage their uploaded documents.



Mobile application / web access

this means the system can be accessed through different devices. Kyuspace is mainly web-based, so students can access and download research documents (pdfs) using phones, tablets, or computers anywhere with internet access.



Access

- the ability of users to reach and use the information system or its resources. In Kyuspace: students and researchers can open the repository online. They can view and download theses, dissertations, and research papers (pdfs).It can be accessed from anywhere using the internet. Example:a student logs into KYU space and downloads a thesis PDF.

DISADVANTAGES OF USER CENTERED

- **High cost** and resources for designing a user-centered system often requires more time, money, and skilled staff. Example: adding features like personalization, feedback systems, and interactive interfaces also the maintenance of the system can be expensive.
- **Complexity in Design Systems** must accommodate different user preferences and behaviors, which can make the design more complicated. Example: Users have different search habits, making it hard to design a “one-size-fits-all” interface.

- **Requires continuous updates** users need changes over time, so the system must be regularly updated to remain effective. Example: Kyuspace would need constant updates to meet students' search habits and new academic requirements.
- **Potential for over-personalization** if a system focuses too much on individual user preferences, it can limit exposure to other useful information. Example: A student may only see research in their department and miss relevant papers from other faculties.
- **User dependence. U**ser-centered systems depend heavily on user input and feedback. If users don't provide feedback or interact properly, the system may not improve or function optimally

CONCLUSION

In conclusion, designing a retrieval system using **LIBRARY MANAGEMENT SYSTEM** has proven that an integrated library system greatly improves the storage, cataloguing, and retrieval of information. By following a systematic design process, the system enhances efficiency and provides remote access to users. The **user-centred approach** further reinforced that a good retrieval system must prioritize user needs through features like searchability, filtering, personalization, and web access. Despite challenges such as high costs and the need for continuous updates, both approaches affirm that an effective retrieval system must balance technical efficiency